Daniyar Abykhanov

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EDUCATION

University of Delaware **Aug 2024 - May 2026**
Master of Science, Data Science *Newark, Delaware*

- **GPA:** 3.7
- **Coursework:** Machine Learning, Introduction to NLP, Algorithm Design and Analysis, Regression Analysis, Applied Multivariate Statistics, Logistic Regression, Database Systems, Principles of Urban Science, Ethics in Data Science and AI

International Informational Technology University **Aug 2020 - May 2024**
Bachelor, Mathematics *Almaty, Kazakhstan*

- **Capstone project:** Neural Networks in Assessing the Disturbance of the Geomagnetic Field

SKILLS

- **Certifications:** Deep Learning Specialization (DeepLearning.AI, Coursera)
- **ML Engineering**: feature engineering, model validation, experiment design(A/B testing), anomaly detection, time series, regression
- **Programming**: Python, SQL, R, SAS
- **ML/LLMs**: scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Transformers, LangChain
- **Data**: Pandas, NumPy, SciPy, PySpark, Matplotlib, Seaborn, Plotly
- **MLOps/Deployment**: Git, Docker, Airflow, FastAPI, MLflow, CI/CD (GitHub Actions), AWS

WORK EXPERIENCE

University of Delaware **Aug 2024 - Present**
Research Assistant *Newark, Delaware*

- Built an end-to-end GAT/GATv2 imputation pipeline in PyTorch Geometric to reconstruct missing values in CMIP-style spatio-temporal data, including graph construction and training/inference workflow
- Designed evaluation across multiple missingness regimes (random/shared/ensemble masks), achieving CE 0.62 to 0.79 (+0.17) vs GCN/GraphConv and ~22% RMSE reduction on held-out regions
- Improved robustness by adding elevation + land/ocean features and seasonal normalization, maintaining performance under 30–70% missingness and reducing seed-to-seed variance by ~35%

Bank CenterCredit **Dec 2022 - Aug 2024**
Machine Learning Engineer *Almaty, Kazakhstan*

- Built and productionized credit scoring models across the lifecycle (origination & collections): origination Gini +10% / KPI +5%, collections Gini +15% / KPI +10%
- Automated end-to-end ML pipeline (data extraction, feature engineering, training/validation, monitoring) with Python, SQL, LightGBM, SHAP, and Airflow, reducing iteration time by 40–60% (about 10 business days to 4–6) and saving 6–10 hours per week
- Implemented data validation and anomaly detection checks to improve reliability, reducing recurring data quality incidents by 30–45% and lowering week-to-week metric variance by 15–25%
- Built a LangChain-based RAG assistant (GPT-3) for call center staff to retrieve up-to-date information from internal knowledge bases, reducing lookup time by 30–50% (about 3–5 minutes to 1–2 minutes per case)
- Developed a separate document summarization workflow to reduce manual review time by 25–45% (about 8–12 minutes to 4–7 minutes per document)

Kazakhtelecom **Jun 2022 – Dec 2022**
Data Science Intern *Almaty, Kazakhstan*

- Built a customer profiling dataset in SQL by joining multiple production tables and engineering key featur es for marketing segmentation and targeting

PROJECTS

AI Agent for Call Center

- Built a LangChain-based RAG assistant for call center staff to retrieve up-to-date information from internal databases and knowledge sources
- Implemented a retrieval workflow to fetch top relevant records and used GPT-3 to generate grounded answers, keeping responses current despite daily data updates
- Reduced average information lookup time by 30–50% (about 3–5 minutes to 1–2 minutes per case) and improved answer usefulness in spot-check reviews from roughly 70% to 85%

Document Summarization

- Built a T5-small–based summarization pipeline to process roughly 5k–20k documents per month (average 2–10 pages), accelerating document handling and knowledge extraction
- Reduced manual review time by 25–45% (for example, 8–12 min to 4–7 min per document) by generating structured summaries